basically an immense expanse of hydrogen. Hydrogen can fuse into helium, and helium has a higher entropy than hydrogen. But for this to happen, it is necessary for a channel to be opened: a star must ignite for hydrogen to begin to burn there into helium. What causes stars to ignite? Another process that increases entropy: the contraction due to gravity of one of the large clouds of hydrogen that sail throughout the galaxy. A contracted cloud of hydrogen has higher entropy than a dispersed one.[97] But the clouds of hydrogen are so vast that they take millions of years to contract. Only after they have become concentrated do they manage to heat up to the point that triggers the process of nuclear fission. The ignition of nuclear fission opens the door that allows the further increase in entropy: hydrogen burning into helium.

The entire history of the universe consists of this halting and leaping cosmic growth of entropy. It is neither rapid nor uniform, because things remain trapped in basins of low entropy (the pile of wood, the cloud of hydrogen . . . ) until something opens a door onto a process that finally allows entropy to increase. The growth of entropy itself happens to open new doors through which entropy can increase further. A dam in the mountains, for instance, retains water until it is gradually worn down over time and the freed water escapes downhill once again, causing entropy to grow. Over the course of this irregular trajectory, large or small portions of the universe remain isolated in relatively stable situations for periods that can be very prolonged.

Living beings are made up of similarly intertwined processes. Photosynthesis deposits low entropy from the sun into plants. Animals feed on low entropy by eating. (If all we needed was energy rather than entropy, we would head for the heat of the Sahara rather than toward our next meal.) Inside every living cell, the complex web of chemical processes is a structure that opens and closes gates through which low entropy can increase. Molecules function as the catalysts that allow the processes to intertwine; or, conversely, they put a brake on them. The increase of entropy in each individual process is what makes the whole thing work. Life is this network of processes for increasing entropy—processes that act as catalysts to each other.[98] It isn’t true, as is sometimes stated, that life generates structures that are particularly ordered, or that locally diminish entropy: it is simply a process that degrades and consumes the low entropy of food; it is a self-structured disordering, no more and no less than in the rest of the universe.

Even the most banal phenomena are governed by the second law of thermodynamics. A stone falls to the ground. Why? One often reads that it’s because the stone places itself “in a state of lower energy” that it ends up lower down. But why does the stone put itself into a state of lower energy? Why should it lose energy if energy is conserved? The answer is that when